A comparative study of bulk RNA-seq deconvolution methods using scRNA-seq references

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SUMMARY OF THE FINAL PROJECT

Title of the project:	*Comparative Evaluation of Immune Cell Deconvolution Methods Using Bulk RNA-seq and Single-Cell RNA-seq Reference Data*
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Abstract

Accurate characterization of the tumor microenvironment (TME) is essential for understanding disease progression and therapeutic response in metastatic prostate adenocarcinoma. While bulk RNA sequencing remains widely used to profile tumor samples, it captures averaged gene expression across heterogeneous cellular populations, limiting direct interpretation of immune cell infiltration. Computational deconvolution methods aim to overcome this limitation by estimating cell-type proportions from bulk transcriptomic data using reference expression profiles derived from single-cell RNA sequencing (scRNA-seq).

This project proposes a comparative evaluation of EPIC, CIBERSORTx, MuSiC, and BisqueRNA applied to bulk RNA-seq data from prostate adenocarcinoma tumors (GSE297742), using scRNA-seq data from metastatic lesions and peripheral blood mononuclear cells (GSE297652) as an external, non sample-matched reference. This cross-cohort setting reflects common real-world scenarios in which scRNA-seq atlases are used to deconvolve independent bulk datasets.

The study aims to assess how preprocessing strategies, gene harmonization, and reference construction influence inferred immune cell proportions within the prostate tumor microenvironment. The central hypothesis is that methodological choices and reference mismatch significantly impact deconvolution robustness and reproducibility. By systematically benchmarking multiple frameworks in a defined clinical context, this work seeks to provide practical guidance for immune profiling in prostate cancer transcriptomic studies.

1. Justification of the Proposal

Prostate adenocarcinoma is one of the most prevalent malignancies in men worldwide and remains a major cause of cancer-related mortality, particularly in advanced and metastatic stages. Increasing evidence highlights the importance of the tumor microenvironment (TME), and especially immune cell infiltration, in shaping tumor progression, therapeutic response, and resistance mechanisms. Reliable characterization of immune composition within prostate tumors is therefore essential for translational oncology research.

Bulk RNA sequencing is widely used to profile tumor transcriptomes due to its scalability and availability in public repositories. However, bulk transcriptomic data represent averaged gene expression across heterogeneous cell populations, obscuring direct interpretation of cellular composition. This limitation is particularly relevant in prostate cancer, where immune infiltration patterns can be heterogeneous and relatively subtle compared to more immunogenic tumor types.

Computational deconvolution methods aim to infer cell-type proportions from bulk RNA-seq data using reference expression profiles derived from purified cells or single-cell RNA sequencing (scRNA-seq). In this project, bulk RNA-seq data from prostate adenocarcinoma tumors (GSE297742) and scRNA-seq data derived from metastatic lesions and peripheral blood mononuclear cells (GSE297652) will be used. Importantly, these datasets are not sample-matched, reflecting a common real-world scenario in which scRNA-seq atlases are used as external references to deconvolve independent bulk cohorts.

While previous studies have benchmarked deconvolution tools, many evaluations rely on simulated mixtures, matched datasets, or idealized reference conditions. Fewer studies explicitly address robustness in cross-cohort settings, where differences in preprocessing pipelines, gene harmonization strategies, and reference construction may substantially influence deconvolution outputs. This project directly addresses this methodological gap by systematically comparing EPIC, CIBERSORTx, MuSiC, and BisqueRNA in a defined metastatic prostate cancer context.

From a scientific perspective, this work contributes to improving reproducibility, methodological transparency, and practical guidance in computational immunogenomics. From a broader perspective, although focused on prostate adenocarcinoma, the methodological insights derived from this study may inform immune deconvolution practices in other solid tumors where non sample-matched scRNA-seq references are commonly used.

2. Personal motivation

My motivation to undertake this Final Master's Project arises from the intersection of my academic background, prior research experience, and long-term interest in computational biology. The topic of immune cell deconvolution was initially proposed by my external supervisor. After conducting an independent review of the relevant literature and exploring current methodological developments in the field, I decided to pursue this line of research due to both its scientific relevance and its strong alignment with my professional goals.

During my internship at the Technical University of Munich, I worked on RNA-seq analysis in the context of tumor-educated platelets, applying statistical frameworks such as edgeR and limma to investigate transcriptomic biomarkers. This experience exposed me to the practical challenges of bulk transcriptomic data, including normalization, differential expression analysis, and biological interpretation of heterogeneous signals.

Beyond its academic value, this project also reflects my broader interest in translational bioinformatics and data-driven healthcare innovation. I am motivated by the challenge of developing reproducible and methodologically sound computational workflows that can ultimately enhance biological interpretation and support precision medicine research.

3. Objectives

To comparatively evaluate immune deconvolution methods in the context of metastatic prostate adenocarcinoma, using non sample-matched single-cell RNA-seq references to infer immune cell composition from bulk RNA-seq tumor profiles, with a focus on methodological robustness and reproducibility within the tumor microenvironment.

To achieve the main objective, the project will pursue the following secondary objectives:

- To characterize the immune landscape of metastatic prostate adenocarcinoma bulk RNA-seq samples (GSE297742) through computational deconvolution.

- To construct and harmonize reference expression profiles derived from scRNA-seq data of metastatic prostate lesions and peripheral blood mononuclear cells (GSE297652).

- To systematically compare EPIC, CIBERSORTx, MuSiC, and BisqueRNA under a cross-cohort reference setting.

- To assess the impact of preprocessing decisions, gene intersection strategies, and reference construction on inferred immune cell proportions within the prostate tumor microenvironment.

- To evaluate concordance and variability across deconvolution frameworks in a clinically relevant metastatic tumor context, rather than under simulated or matched conditions.

- To derive methodological recommendations for immune profiling in prostate cancer transcriptomic studies and similar solid tumor settings.

4. Hypothesis

Immune cell deconvolution results derived from bulk RNA-seq data are significantly influenced by preprocessing strategies and reference construction choices in single-cell RNA-seq datasets, leading to measurable variability across computational methods.

To test the main hypothesis, the project will address the following research questions:

1. How do different single-cell RNA-seq preprocessing strategies (e.g., quality control thresholds, normalization approaches, gene filtering) affect the construction of reference signatures?

2. To what extent does gene harmonization between bulk and single-cell datasets influence deconvolution outputs?

3. Do different deconvolution algorithms (e.g., EPIC, CIBERSORTx, MuSiC, BisqueRNA) produce consistent immune cell proportion estimates under standardized preprocessing conditions?

4. How sensitive are deconvolution results to variations in reference construction and gene selection strategies?

5. Can methodological standardization reduce variability across methods and improve reproducibility of inferred immune cell proportions?

5. Impact on Sustainability, Socio-Ethical, and Diversity

This Final Master's Project is framed within the transversal competence of Ethical and Global Commitment (CCEG) promoted by the Universitat Oberta de Catalunya (UOC) (Universitat Oberta de Catalunya, 2022). This competence encourages academic and professional practice that is ethical, socially responsible, sustainable, and respectful of diversity and human rights. The present project integrates these dimensions throughout its conceptual design, methodological framework, and data usage strategy, in alignment with the United Nations Sustainable Development Goals (SDGs) (United Nations, 2015).

5.1 Sustainability

From a sustainability perspective, this project contributes primarily to:

- SDG 3 – Good Health and Well-being
- SDG 9 – Industry, Innovation and Infrastructure

The study focuses on metastatic prostate adenocarcinoma, a clinically significant disease with substantial global health impact. By improving methodological robustness in immune cell deconvolution, the project supports more accurate interpretation of tumor transcriptomic data, which may contribute indirectly to better understanding of tumor–immune interactions and future translational research efforts. In this sense, the project aligns with SDG 3 by promoting advancements in biomedical knowledge that may inform improved diagnostic and therapeutic strategies.

Furthermore, the project contributes to SDG 9 by advancing computational innovation in biomedical data analysis. The development of reproducible and robust computational workflows strengthens research infrastructure and promotes responsible digital transformation in health research.

Importantly, this study exclusively relies on publicly available transcriptomic datasets retrieved from the NCBI Gene Expression Omnibus (GEO). By reusing existing anonymized patient data rather than generating new experimental sequencing data, the project promotes sustainable research practices, minimizing additional resource consumption and laboratory costs while maximizing the scientific value of publicly funded datasets.

5.2 Ethical Behaviour and Social Responsibility

The project adheres strictly to ethical standards in biomedical data analysis. All datasets used (GSE297742 and GSE297652) are publicly available and anonymized, with no personally identifiable information accessible. As such, the study complies with data protection principles consistent with the General Data Protection Regulation (GDPR).

The research does not involve direct patient recruitment, experimental interventions, or access to confidential clinical records. Instead, it relies on secondary analysis of ethically approved and publicly shared transcriptomic data.

Methodological transparency is also an ethical commitment. By clearly documenting preprocessing steps, harmonization strategies, and analytical decisions, the project promotes reproducibility and accountability in computational research, reducing the risk of opaque or irreproducible findings. This aligns directly with the CCEG principles of honesty, responsibility, and professional integrity (Universitat Oberta de Catalunya, 2022).

5.3 Diversity and Human Rights

Biomedical transcriptomic datasets may reflect demographic imbalances, including potential underrepresentation of certain age groups, ethnicities, or clinical subtypes. Although prostate adenocarcinoma is a male-specific disease, variability in ethnicity, disease stage, and treatment background may still influence transcriptomic profiles.

Where metadata are available, the project will acknowledge the demographic composition of the cohorts and explicitly discuss potential biases or limitations. If demographic balance cannot be assessed due to incomplete metadata, this limitation will be transparently reported.

Recognizing these limitations is essential to avoid overgeneralization and to ensure responsible interpretation of computational immune profiling results. This approach aligns with the CCEG dimension related to diversity and human rights and supports inclusive research practices consistent with the SDGs (United Nations, 2015).

6. Methodology

6.1 Research Strategy

This project follows a quantitative empirical research strategy within the field of computational bioinformatics. More specifically, it adopts a structured benchmarking framework aimed at evaluating the robustness of existing immune deconvolution methods under realistic cross-cohort conditions.

The study does not generate new biological data. Instead, it performs secondary analysis of publicly available bulk RNA-seq and single-cell RNA-seq datasets retrieved from the NCBI Gene Expression Omnibus (GEO). The bulk and single-cell datasets are not patient-matched. While this prevents direct biological validation at the individual level, it reflects a common translational research scenario in which publicly available single-cell atlases are used as external references for independent bulk tumor cohorts.

This design allows evaluation of methodological robustness in a context that closely resembles real-world applications of immune deconvolution.

6.2 Analytical Framework

The analytical pipeline is implemented in a modular and reproducible R Markdown workflow. The pipeline is organized into clearly separated stages to ensure transparency, traceability, and methodological consistency.

Stage 1: Dataset Acquisition and Initial Assessment

Two publicly available datasets are used:

- Bulk RNA-seq data from metastatic prostate adenocarcinoma tumors (GSE297742).

- Single-cell RNA-seq data from metastatic prostate cancer lesions and peripheral blood mononuclear cells (GSE297652).

Metadata and gene annotations are evaluated to ensure compatibility between datasets and to define harmonization strategies prior to deconvolution.

Stage 2: Single-Cell RNA-seq Preprocessing and Annotation

Single-cell preprocessing follows established best-practice recommendations (Luecken & Theis, 2019; Heumos et al., 2023). The workflow includes:

- Quality control filtering

- Normalization using SCTransform

- Dimensionality reduction
- Clustering
- Marker gene identification

Cluster annotation is performed using a semi-structured strategy based on canonical immune markers and cluster-level expression scoring. Clusters are assigned cell-type labels with defined confidence levels, allowing exclusion of uncertain clusters prior to reference construction. This approach ensures traceability and reduces subjectivity in reference definition.

Stage 3: Reference Signature Construction

A cell-type reference expression matrix is constructed by averaging normalized expression values per annotated cell type. Only confidently annotated clusters are retained.

This matrix represents tumor and immune cell populations relevant to metastatic prostate cancer and serves as the input reference for all downstream deconvolution analyses.

Stage 4: Preliminary Technical Validation via Pseudobulk

Before applying deconvolution methods to independent bulk tumor samples, controlled pseudobulk mixtures are generated from the annotated single-cell dataset.

These simulated mixtures contain known cell-type proportions and serve as a technical validation step to:

- Verify correct implementation of each method
- Assess basic recovery capacity under controlled conditions

This step functions as a methodological consistency check and does not replace biological validation.

Stage 5: Bulk RNA-seq Harmonization

Bulk RNA-seq data are preprocessed and harmonized with the single-cell reference matrix. Gene intersection strategies and normalization compatibility are explicitly evaluated to minimize bias introduced by gene annotation discrepancies.

Gene matching procedures are transparently documented within the workflow.

Stage 6: Implementation of Deconvolution Methods

Selected immune deconvolution frameworks are applied under standardized preprocessing conditions:

- EPIC (R implementation)
- MuSiC (R implementation)
- CIBERSORTx (web-based implementation)
- BisqueRNA (if computationally feasible)

Input matrices are derived from the harmonized reference and bulk datasets to ensure comparability across methods.

Stage 7: Comparative and Robustness Evaluation

The comparative framework focuses on methodological robustness rather than clinical validation. Specifically, the analysis evaluates:

- Concordance of inferred immune cell proportions across methods
- Sensitivity to preprocessing decisions
- Impact of gene filtering and reference construction choices
- Variability of outputs under cross-cohort conditions

Quantitative comparison includes correlation analyses between methods and assessment of variability patterns across analytical settings.

6.3 Reproducibility and Transparency

All analytical steps are implemented in R Markdown and version-controlled using Git. The workflow is modular, with clearly separated stages for preprocessing, reference construction, validation, and deconvolution.

Explicit documentation of parameters, filtering criteria, and harmonization strategies ensures methodological transparency and reproducibility.

7. Planning

7.1 Planning Strategy

The project follows an iterative and modular planning strategy aligned with the implemented analytical pipeline. Although the stages follow a logical dependency structure, the workflow allows refinement and adjustment of preprocessing parameters and reference construction decisions based on intermediate results.

The planning is organized around analytical milestones rather than isolated tasks, ensuring progressive validation and integration of results throughout the development process.

Estimated workload: 15–20 hours per week.

Phase 1 – Conceptual Framework and Literature Grounding

Timeline: Month 1

- Literature review on immune deconvolution methods.
- Definition of benchmarking criteria.
- Finalization of hypothesis and objectives.
- Selection and exploration of candidate datasets.

Milestone: Conceptual and methodological framework finalized.

Dependency: None.

Phase 2 – Single-Cell Processing and Reference Construction

Timeline: Months 1–2

- Quality control and normalization of scRNA-seq data.

- Clustering and marker identification.
- Semi-structured cell-type annotation.
- Construction of reference signature matrix.

Milestone: Final reference matrix generated and validated.

Dependency: Phase 1.

Potential risk:

- Ambiguous cluster annotation.
  Mitigation:
- Use of canonical markers and confidence-level filtering.

Phase 3 – Preliminary Technical Validation (Pseudobulk)

Timeline: Month 2

- Generation of pseudobulk mixtures.
- Implementation of selected deconvolution methods.
- Basic recovery assessment under controlled conditions.

Milestone: Technical validation completed.

Dependency: Phase 2.

Potential risk:

- Inconsistent method behavior due to parameter settings.
  Mitigation:
- Standardized input formatting and parameter documentation.

Phase 4 – Bulk RNA-seq Harmonization and Cross-Cohort Application

Timeline: Months 2–3

- Preprocessing and normalization of bulk RNA-seq data.
- Gene intersection and harmonization with reference matrix.
- Application of deconvolution methods to real bulk tumor samples.

Milestone: First complete cross-cohort deconvolution results.

Dependency: Phase 2 (reference matrix required).

Potential risk:

- Gene identifier mismatches.
  Mitigation:
- Explicit gene mapping and controlled filtering strategies.

Phase 5 – Comparative and Robustness Analysis

Timeline: Months 3–4

- Cross-method comparison.
- Sensitivity analysis to preprocessing and reference construction.
- Assessment of variability patterns under cross-cohort conditions.

Milestone: Robustness analysis completed.

Dependency: Phases 3 and 4.

Phase 6 – Writing, Integration, and Revision

Timeline: Months 3–5 (overlapping)

- Drafting of methods and results sections.
- Integration of figures and tables.
- Iterative revision based on supervisor feedback.

Milestone: Complete draft of thesis.

Dependency: Progressive completion of analytical stages.

7.3 Risk Assessment and Contingency Plan

The main methodological risks include:

- Non-matched dataset limitations.
- Sensitivity to reference construction choices.
- Computational inconsistencies across methods.
- Incomplete metadata availability.

These risks are mitigated through:

- Explicit acknowledgment of cross-cohort design.
- Preliminary pseudobulk validation.
- Standardized preprocessing procedures.
- Transparent documentation of assumptions and parameters.

7.4 Alignment with Academic Deliverables

The planning is aligned with the formal evaluation milestones (M1–M3):

- M1: Conceptual framework and methodological design.
- M2: Implementation and intermediate analytical results.
- M3: Final comparative evaluation and full thesis integration.

Gantt Chart

Each block ▇ is roughly equivalent to one part of the month

Task	M1	M2	M3	M4	M5	M6
Literature review & framework	■■	■■				
Dataset acquisition & exploration	■■					
scRNA preprocessing & clustering	■■	■■				
Reference construction		■■	■			
Pseudobulk validation		■■				
Bulk harmonization		■	■■			
Deconvolution methods implementation		■	■■	■		
Robustness & sensitivity analysis			■■	■■		
Interpretation & integration				■■	■	
Writing (progressive)	■	■■	■■	■■	■	
Revision & final adjustments					■■	■

The project begins with a conceptual and methodological grounding phase (M1), ensuring that benchmarking criteria and dataset selection are clearly defined before intensive computational analysis begins.

Single-cell preprocessing and reference construction (M1–M2) represent a foundational stage, as all subsequent analyses depend on the quality and reliability of the constructed reference matrix.

Preliminary pseudobulk validation (M2) functions as an internal methodological checkpoint before proceeding to real-world cross-cohort application.

Bulk harmonization and deconvolution implementation (M2–M3) form the core analytical phase of the project. Robustness and sensitivity analyses (M3–M4) extend this phase by evaluating methodological stability.

Writing activities overlap progressively with analytical stages to ensure early integration of results and reduce final-stage workload concentration.

Final revisions and integration (M5–M6) focus on consolidating the methodological narrative, discussing limitations, and ensuring coherence across sections.

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